AWS Whitepaper

Overview of Deployment Options on AWS



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Overview of Deployment Options on AWS: AWS Whitepaper

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Publication date: **May 31, 2024** (*Document Revisions*)

Abstract

Amazon Web Services (AWS) offers multiple options for provisioning infrastructure and deploying your applications. Whether your application architecture is a simple three-tier web application or a complex set of workloads, AWS offers deployment services to meet the requirements of your application and your organization.

This whitepaper is intended for individuals looking for an overview of the different deployment services offered by AWS. It lays out common features available in these deployment services, and articulates basic strategies for deploying and updating application stacks.

Introduction

Designing a deployment solution for your application is a critical part of building a well-architected application on AWS. Based on the nature of your application and the underlying services that it requires, you can use AWS services to create a flexible deployment solution that can be tailored to fit the needs of both your application and your organization.

The constantly growing catalog of AWS services not only complicates the process of deciding which services will compose your application architecture, but also the process of deciding how you will create, manage, and update your application. When designing a deployment solution on AWS, you should consider how your solution will address the following capabilities:

- **Provision** - Create the raw infrastructure or managed service infrastructure required for your application.
- **Configure** - Customize your infrastructure based on environment, runtime, security, availability, performance, network or other application requirements.
- **Deploy** - Install or update your application components onto infrastructure resources and manage the transition from a previous application version to a new application version.
- **Scale** - Proactively or reactively adjust the amount of resources available to your application based on a set of user-defined criteria.
- **Monitor** - Provide visibility into the resources that are launched as part of your application architecture. Track resource usage, deployment success or failure, application health, application logs, configuration drift, and more.

This whitepaper highlights the deployment services offered by AWS and outlines strategies for designing a successful deployment architecture for any type of application.

AWS Deployment Services

The task of designing a scalable, efficient, and cost-effective deployment solution should not be limited to how you will update your application version, but should also consider how you will manage supporting infrastructure throughout the complete application lifecycle. Resource provisioning, configuration management, application deployment, software updates, monitoring, access control, and other concerns are all important factors to consider when designing a deployment solution.

AWS services can provide management capabilities for one or more aspects of your application lifecycle. Depending on your desired balance of control (manual management of resources) versus convenience (AWS management of resources) and the type of application, these services can be used on their own or combined to create a feature-rich deployment solution. This section will provide an overview of the AWS services that can be used to enable organizations to more rapidly and reliably build and deliver applications.

AWS CloudFormation

AWS CloudFormation is a service that enables customers to provision and manage almost any AWS resource using a custom template language expressed in YAML or JSON. An CloudFormation template creates infrastructure resources in a group called a *stack*, and allows you to define and customize all components needed to operate your application while retaining full control of these resources. Using templates introduces the ability to implement version control on your infrastructure, and the ability to quickly and reliably replicate your infrastructure.

CloudFormation offers granular control over the provisioning and management of all application infrastructure components, from low-level components such as route tables or subnet configurations, to high-level components such as CloudFront distributions. CloudFormation is commonly used with other AWS deployment services or third-party tools, combining CloudFormation with more specialized deployment services to manage deployments of application code onto infrastructure components.

AWS offers extensions to the CloudFormation service in addition to its base features:

- AWS Cloud Development Kit (AWS CDK) is an open source software development kit (SDK) to programmatically model AWS infrastructure with TypeScript, JavaScript, Python, Java, or C#/.NET.

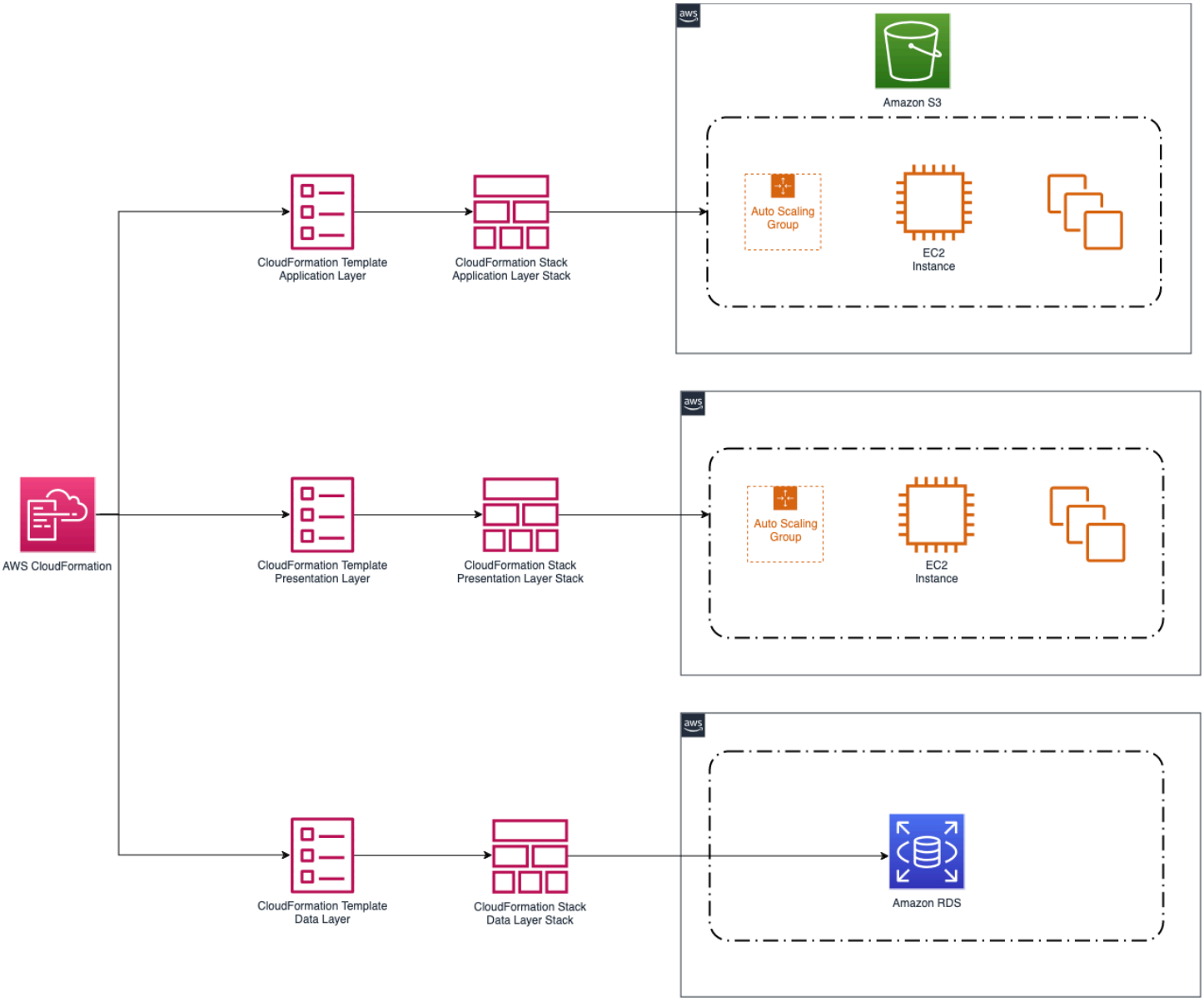
- AWS Serverless Application Model (AWS SAM) is an open source framework to simplify building serverless applications on AWS. It provides shorthand syntax to express functions, APIs, databases, and event source mappings.

Table 1: AWS CloudFormation deployment features

Capability	Description
Provision	CloudFormation will automatically create and update infrastructure components that are defined in a template. Refer to AWS CloudFormation Best Practices for more details on creating infrastructure using CloudFormation templates.
Configure	CloudFormation templates offer extensive flexibility to customize and update all infrastructure components. Refer to CloudFormation Template Anatomy for more details on customizing templates.
Deploy	Update your CloudFormation templates to alter the resources in a stack. Depending on your application architecture, you might need an additional deployment service to update the application version running on your infrastructure. Refer to Deploying Applications on Amazon EC2 with AWS CloudFormation for more details on how CloudFormation can be used as a deployment solution.
Scale	CloudFormation will not automatically handle infrastructure scaling on your behalf; however,

Capability	Description
	you can configure auto scaling policies for your resources in a CloudFormation template.
Monitor	CloudFormation provides native monitoring of the success or failure of updates to infrastructure defined in a template, as well as *drift detection* to monitor when resources defined in a template do not meet specifications. Additional monitoring solutions will need to be in place for application-level monitoring and metrics. Refer to Monitoring the Progress of a Stack Update for more details on how CloudFormation monitors infrastructure updates.

The following diagram shows a common use case for CloudFormation. Here, CloudFormation templates are created to define all infrastructure components necessary to create a simple three-tier web application. In this example, we are using bootstrap scripts defined in CloudFormation to deploy the latest version of our application onto Amazon EC2 instances; however, it is also a common practice to combine additional deployment services with CloudFormation (using CloudFormation only for its infrastructure management and provisioning capabilities). Note that more than one CloudFormation template is used to create the infrastructure. In the diagram, CloudFormation is used to create all infrastructure components including IAM roles, VPCs, subnets, route tables, security groups, and Amazon S3 bucket policies. Separate CloudFormation templates are used to build each domain of the application architecture.



AWS CloudFormation use case

AWS Elastic Beanstalk

AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, .NET Core, PHP, Node.js, Python, Ruby, Go, or Docker on familiar servers such as Apache, Nginx, Passenger, and IIS. Elastic Beanstalk is a complete application management solution, and manages all infrastructure and platform tasks on your behalf.